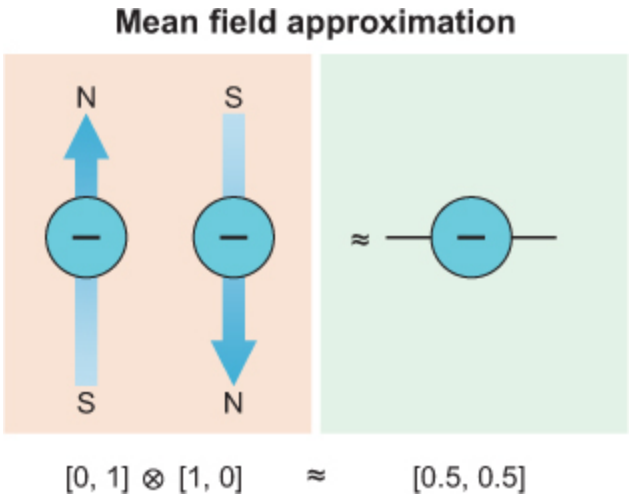
Figure 9.19. The joint action for a pair of electron spins is the outer product between their individual action vectors, which is a 4-element one-hot vector. Rather than using this exact joint action, we can approximate it by taking the average of these two action vectors, resulting in what's called the mean field approximation. For two electrons together, with one spin-up and the other spin-down, the mean field approximation results in reducing this two electron system to a single "virtual" electron with an indeterminate spin of [0.5,0.5].



This means that our mean field vector for each agent will only be a 2-element vector for the 1D Ising model and also for the 2D and higher dimensional Ising models. Our environment can be arbitrarily complex and high-dimensional, and it will still be computationally easy.

Let's see how mean field Q-learning (MF-Q) works on the 2D Ising model. The 2D Ising model is exactly the same as the 1D version, except now it's a 2D grid (i.e., a matrix). The agent in the top-left corner will have its left neighbor be the agent in the top-right corner, and its neighbor above will be the agent in the bottom-left corner, so the grid is actually wrapped around the surface of a sphere (figure 9.20).

Figure 9.20. We represent the 2D Ising model as a 2D square grid (i.e., a matrix), but we design it so that there are no boundaries, and the agents that appear to be on a boundary are actually immediately adjacent to the agents on the opposite side of the grid. Thus the 2D grid is really a 2D grid wrapped around the surface of a sphere.